Construct Binary Tree from Preorder and Inorder Traversal (medium)

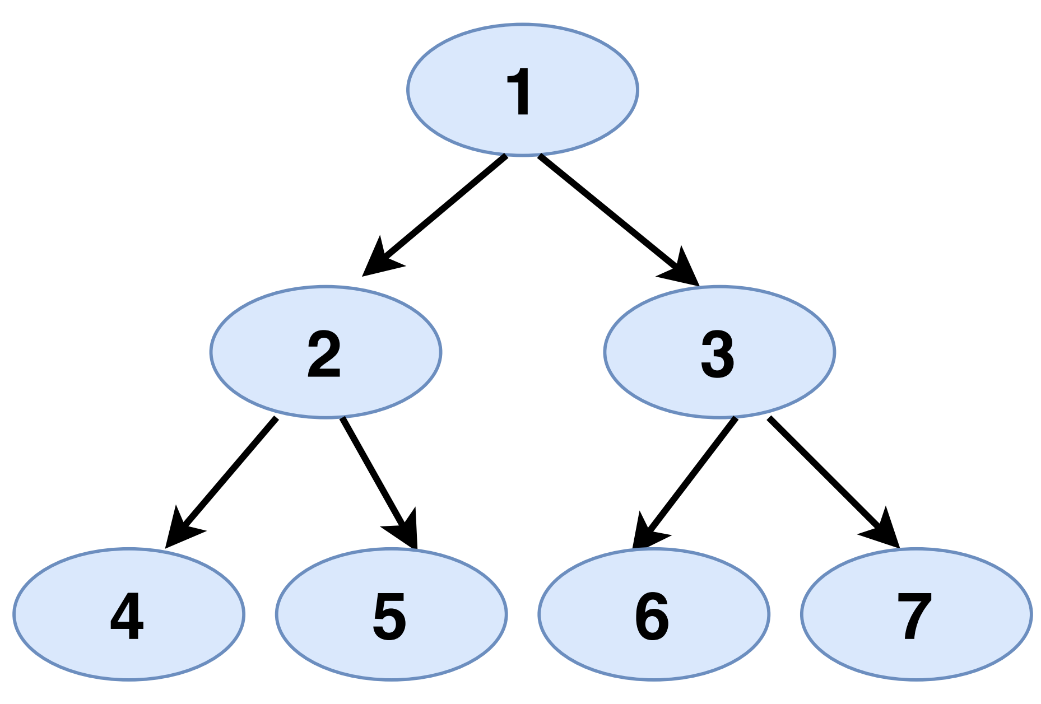
Problem Statement

Given the preorder and inorder traversal sequences of a binary tree, your task is to reconstruct this binary tree. Assume that the tree does not contain duplicate values.

Example Generation

Example 1:

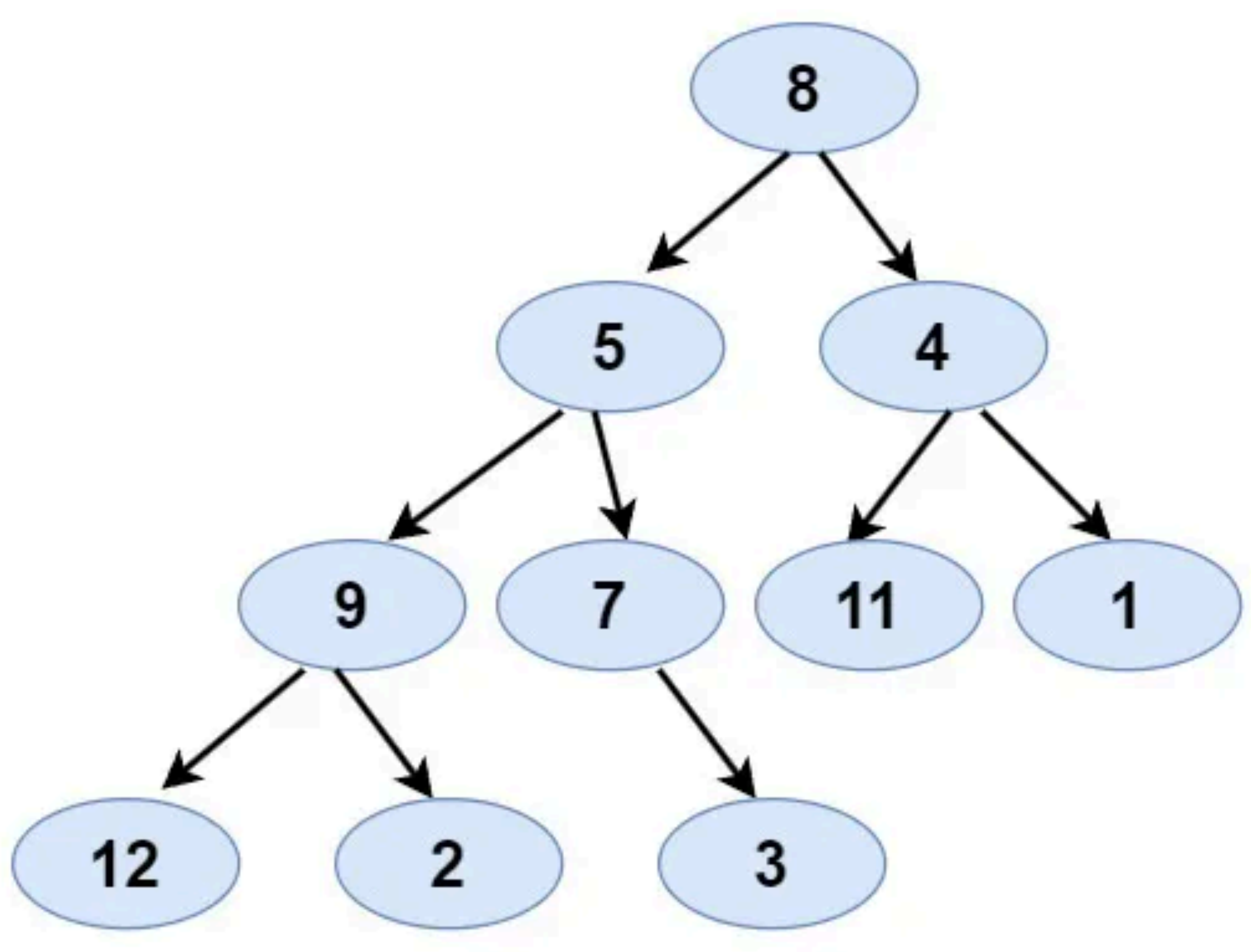
- Input:
  - Preorder: [1,2,4,5,3,6,7]
  - Inorder: [4,2,5,1,6,3,7]
- Expected Output:
  - Tree Representation: [1,2,3,4,5,6,7]



- Justification:
  - The first value in preorder (1) is the root. In the inorder list, everything left of value 1 is the left subtree and everything on the right is the right subtree. Following this pattern recursively helps in reconstructing the binary tree. All `null` value represents the leaf node.

Example 2:

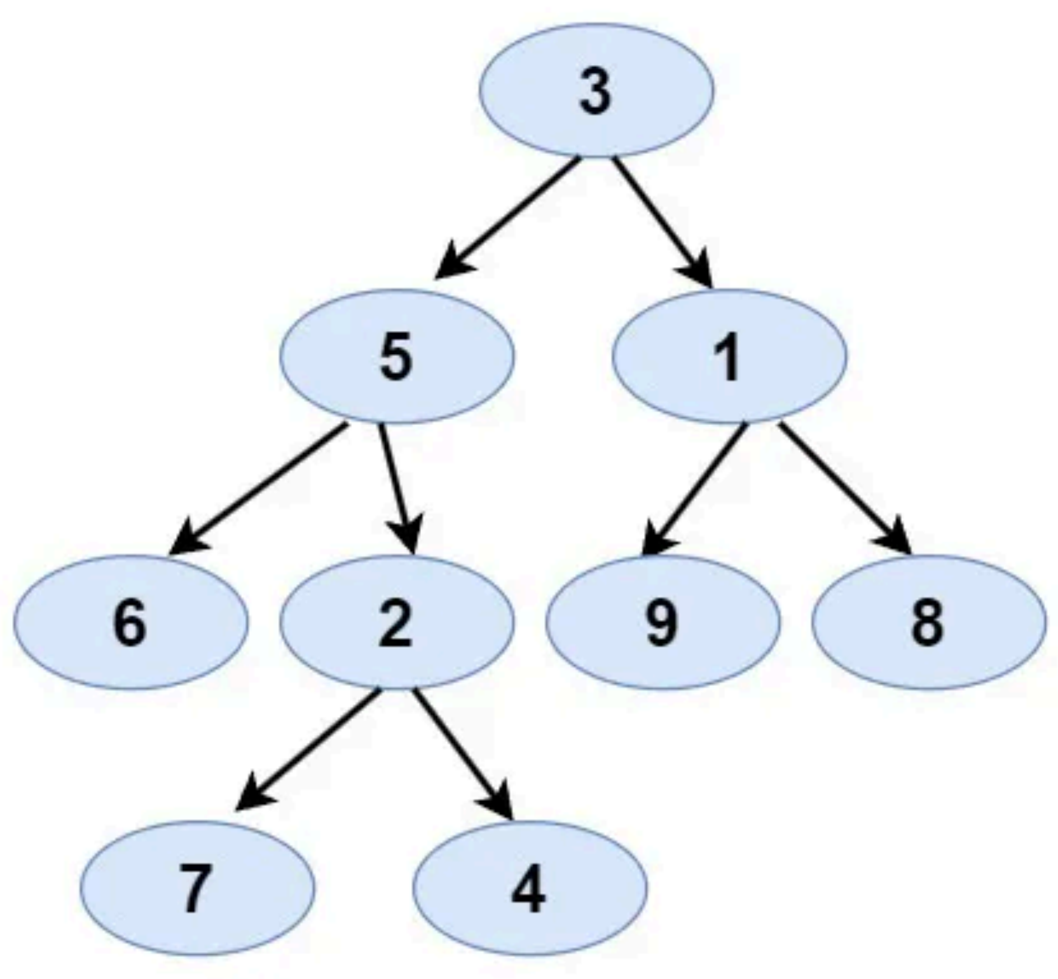
- Input:
  - Preorder: [8,5,9,7,1,12,2,4,11,3]
  - Inorder: [9,5,1,7,2,12,8,4,3,11]
- Expected Output:
  - Tree Representation: [8,5,4,9,7,11,1,12,2,null,3]



- Justification:
  - Start with 8 (from preorder) as the root. Splitting at 8 in inorder, we find the left and right subtrees. Following this pattern recursively, we can construct the tree.

Example 3:

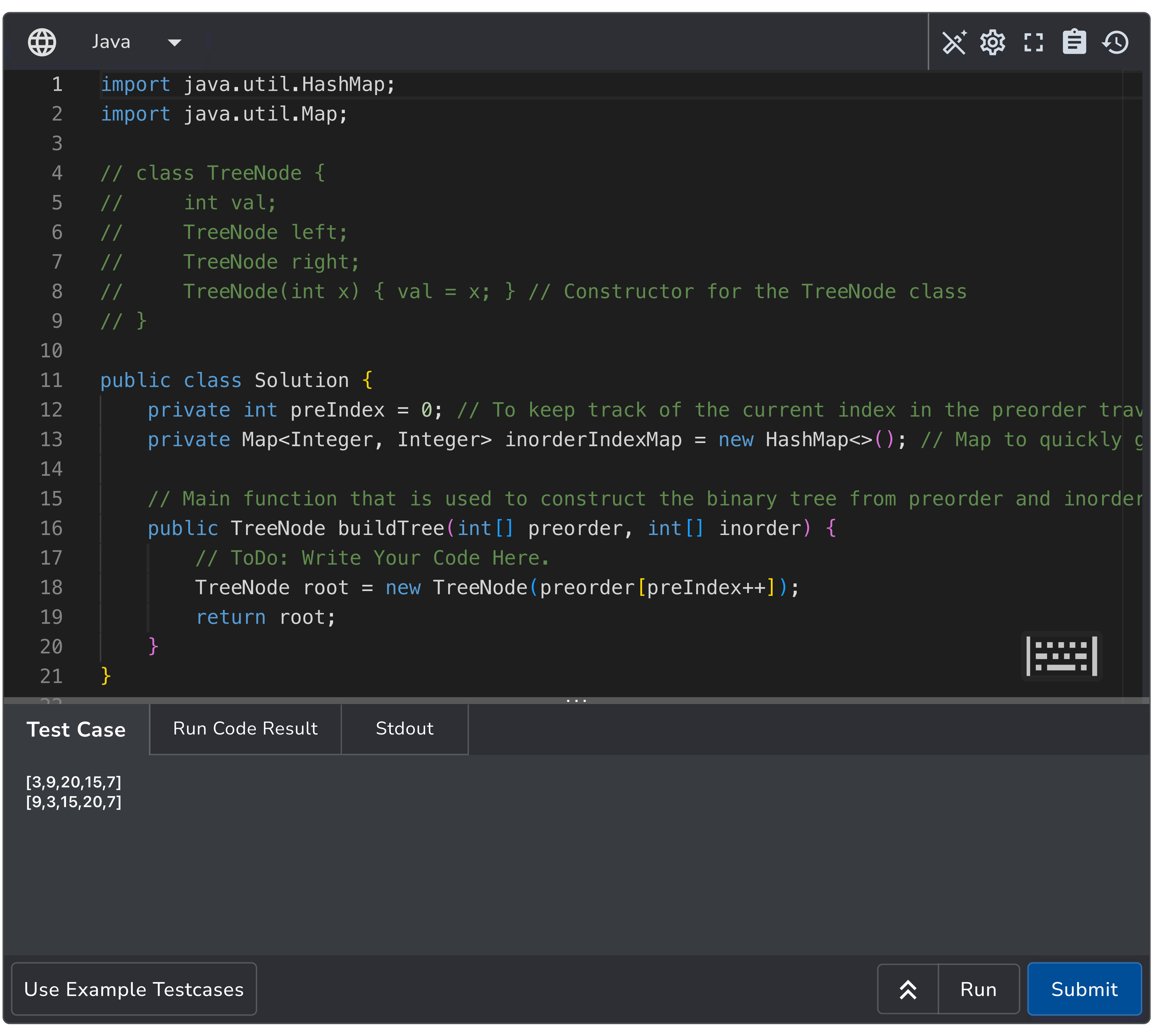
- Input:
  - Preorder: [3,5,6,2,7,4,1,9,8]
  - Inorder: [6,5,7,2,4,3,9,1,8]
- Expected Output:
  - Tree Representation: [3,5,1,6,2,9,8,null,null,7,4]



- Justification:
  - Following the same approach, using 3 as root from preorder, we split the inorder sequence into left and right subtrees and continue recursively.

Try it yourself

Try solving this question here:



```java
import java.util.HashMap;
import java.util.Map;

// class TreeNode {
//     int val;
//     TreeNode left;
//     TreeNode right;
//     TreeNode(int x) { val = x; } // Constructor for the TreeNode class
// }

public class Solution {
    private int preIndex = 0; // To keep track of the current index in the preorder trav
    private Map<Integer, Integer> inorderIndexMap = new HashMap<>(); // Map to quickly g

    // Main function that is used to construct the binary tree from preorder and inorder
    public TreeNode buildTree(int[] preorder, int[] inorder) {
        // ToDo: Write Your Code Here.
        TreeNode root = new TreeNode(preorder[preIndex++]);
        return root;
    }
}
```

```
[3,9,20,15,7]
[9,3,15,20,7]
```

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Solution: Validate Binary Search Tree

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Solution: Construct Binary Tree From Preorder And Inorder Traversal

☐ Mark as Completed